

mcnutt-et-al-2020-surfacing-visualization-mirages

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Not available for this paper.

Abstract

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Dirty data and deceptive design practices can undermine, invert, or invalidate the purported messages of charts and graphs. These failures can arise silently: a conclusion derived from a particular visualization may look plausible unless the analyst looks closer and discovers an issue with the backing data, visual specification, or their own assumptions. We term such silent but significant failures visualization mirages. We describe a conceptual model of mirages and show how they can be generated at every stage of the visual analytics process. We adapt a methodology from software testing, metamorphic testing, as a way of automatically surfacing potential mirages at the visual encoding stage of analysis through modifications to the underlying data and chart specification. We show that metamorphic testing can reliably identify mirages across a variety of chart types with relatively little prior knowledge of the data or the domain.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.